Project: Retail Sales Analysis (2014–2017)

Tools Used: SQL • Power BI

A regional retail company has experienced consistent revenue growth over four years but has raised concerns that profitability is not keeping pace. The objective is to determine where the business is generating value, where it is losing money, and what actions are supported by the data.

Business Questions

- What are total sales, profit, and order volume, and what is the overall profit margin?
- Which regions and states are driving the most revenue and profit?
- Which product categories and subcategories are the most and least profitable?
- How do Consumer, Corporate, and Home Office segments compare on sales and profit?
- How have sales and profit trended over time, and are there seasonal patterns?

Tools & Approach

I used SQL to aggregate and calculate key metrics including total revenue, profit, profit margin, and order volume by category, region, and time period and then built a three-page Power BI dashboard to visualize the findings and enable interactive filtering by year, category, and region.

Dashboard Page 1: Trend Analysis

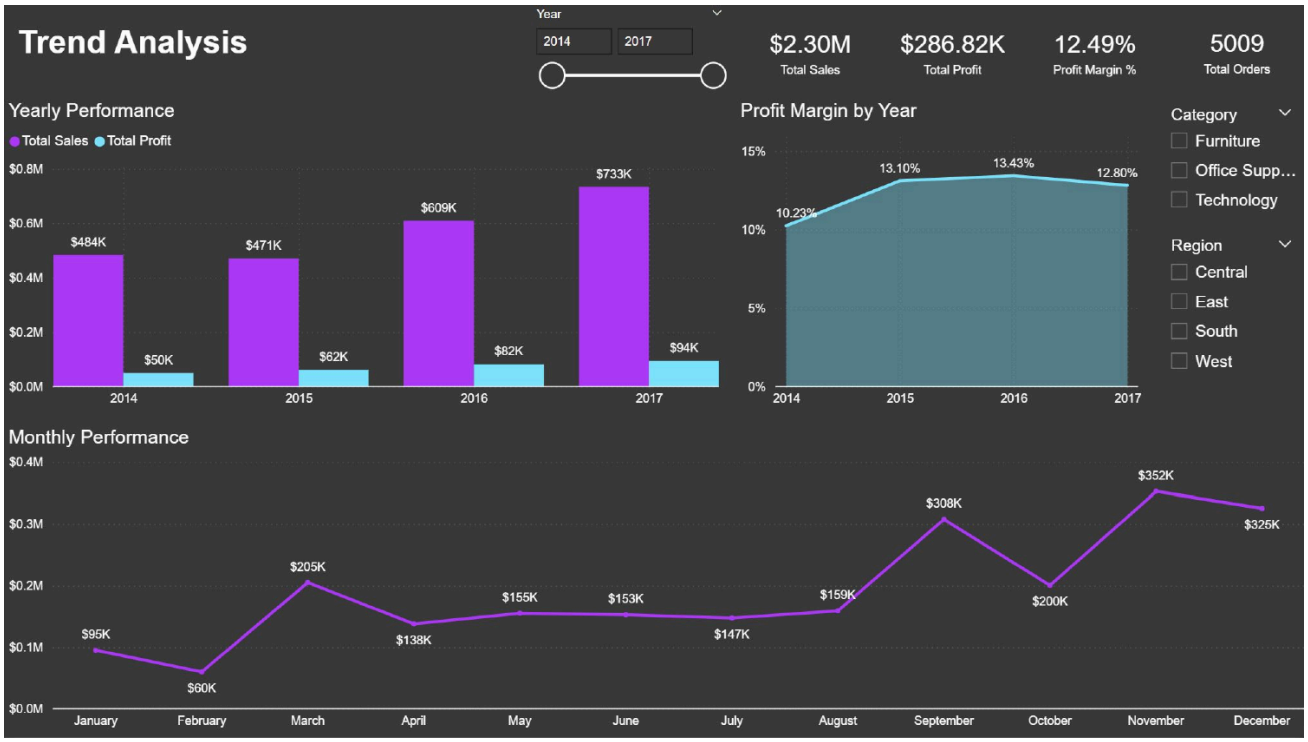


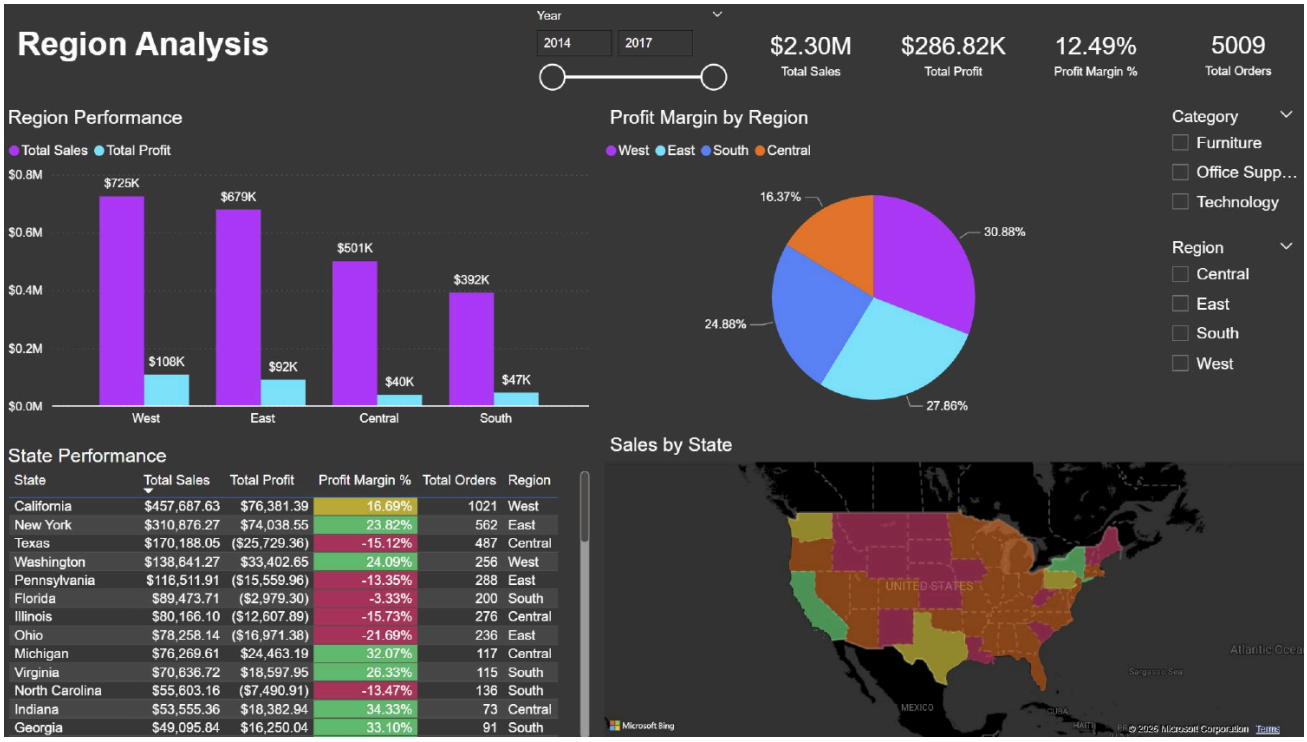
Figure 1 — Trend Analysis dashboard showing yearly and monthly sales and profit margin performance

Key Findings

- Total sales grew from $484K in 2014 to $733K in 2017, with profit rising from $50K to $94K — consistent growth across all four years.

- Profit margin peaked at 13.43% in 2016 before dipping to 12.80% in 2017. Revenue is growing, but the margin decline signals rising costs or discounting worth investigating.
- Monthly sales show two clear peaks: March ($205K) and a strong fall/holiday surge from September through November, peaking at $352K in November.
- January and February are the lowest months at $95K and $60K respectively. These slow months warrant a deeper investigation.

Dashboard Page 2: Region Analysis



State	Total Sales	Total Profit	Profit Margin %	Total Orders	Region
California	$457,687.63	$76,381.39	16.69%	1021	West
New York	$310,876.27	$74,038.55	23.82%	562	East
Texas	$170,188.05	($25,729.36)	-15.12%	487	Central
Washington	$138,641.27	$33,402.65	24.09%	256	West
Pennsylvania	$116,511.91	($15,559.96)	-13.35%	288	East
Florida	$89,473.71	($2,979.30)	-3.33%	200	South
Illinois	$80,166.10	($12,607.89)	-15.73%	276	Central
Ohio	$78,258.14	($16,971.38)	-21.69%	236	East
Michigan	$76,269.61	$24,463.19	32.07%	117	Central
Virginia	$70,636.72	$18,597.95	26.33%	115	South
North Carolina	$55,603.16	($7,490.91)	-13.47%	136	South
Indiana	$53,555.36	$18,382.94	34.33%	73	Central
Georgia	$49,095.84	$16,250.04	33.10%	91	South

Figure 2 — Region Analysis dashboard showing performance by region and state with geographic mapping

Key Findings

- The West region leads with $725K in sales and $108K in profit. California is the top state overall at $457K in revenue and $76K in profit.
- New York is the most efficient state in the dataset: $310K in sales and $74K in profit — nearly equal to California's profit at less than half the order volume (562 vs. 1,021 orders). This suggests significant untapped growth potential.
- The Central region is a financial concern. Texas (-15.12% margin), Illinois (-15.73%), and Ohio (-21.69%) are all losing money. This warrants a deeper look at product mix, pricing, and cost structure in that region before expanding further.
- Washington and Virginia both show strong margins (24% and 26%), indicating that geographic profitability is uneven even within the same region.

Dashboard Page 3: Product Performance Analysis

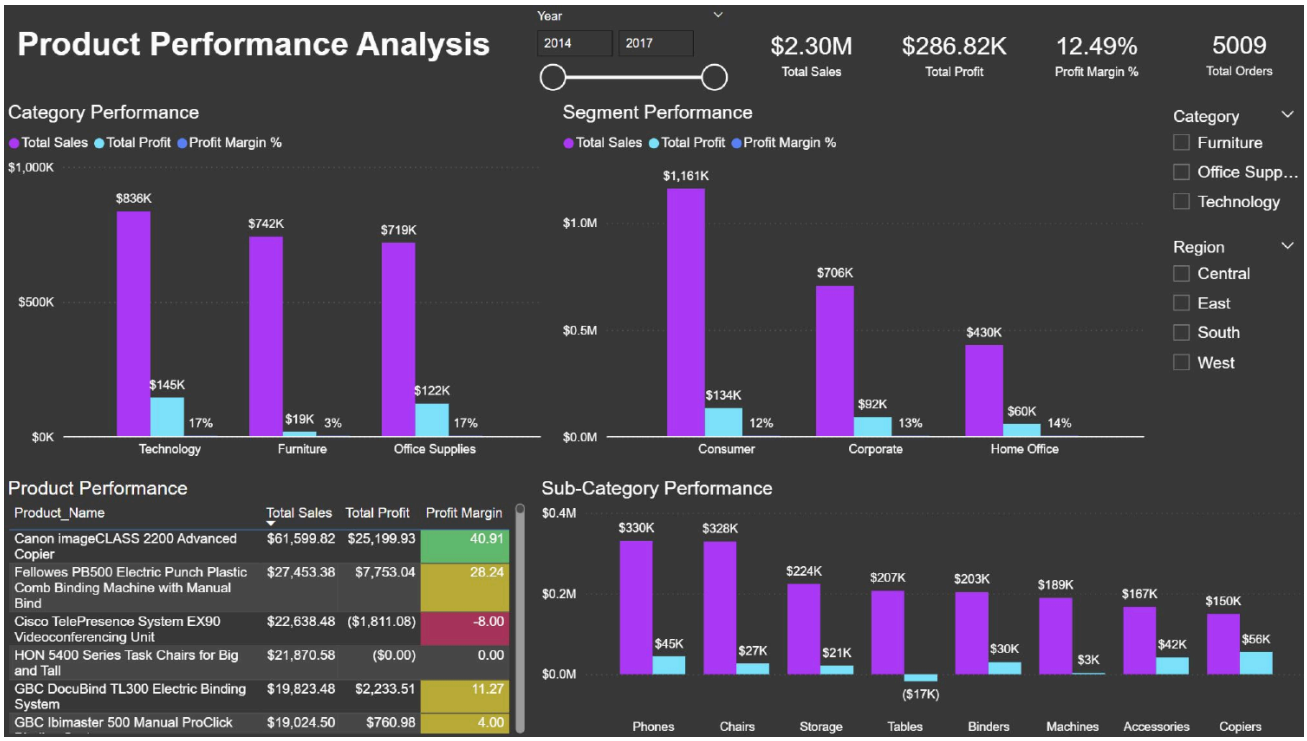


Figure 3 — Product Performance Analysis showing category, segment, subcategory, and top product breakdown

Key Findings

- Technology is the strongest category with $836K in revenue, $145K in profit, and a 17% margin. Office Supplies also holds a 17% margin at lower volume.
- Furniture is a hidden problem: $742K in revenue but only a 3% profit margin ($19K profit). High sales volume does not mean profitability, and this category deserves a cost and pricing review.
- The Consumer segment generates $1.16M in revenue — nearly double Corporate ($706K). Consumers should remain the primary focus of sales and marketing strategy.
- Copiers are the standout subcategory by profitability. The Canon imageCLASS 2200 Advanced Copier leads all products with a 40.91% profit margin on $61.6K in sales, making it the single best-performing product in the dataset.
- Tables generate negative profit (-$17K shown in the subcategory chart), making them the weakest subcategory and a candidate for pricing or sourcing review.

Summary & Recommendations

The business is growing in revenue and profit, but the 2017 margin dip and concentrated losses in the Central region and Furniture category signal areas that need attention. The following are the highest-priority takeaways:

- Investigate the 2017 margin decline before assuming overall business health is strong.
- Develop a growth strategy for New York, which is highly efficient and likely underserved.
- Conduct a full review of Central region operations — three of the top states by order volume are losing money.
- Reassess Furniture pricing and cost structure. $742K in sales at 3% margin is a significant opportunity cost.

- Double down on Technology and Copiers, which are the clearest margin drivers in the portfolio.
- Review seasonal strategy for Q1. January and February are consistently slow; the goal should be understanding why.

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